Tonmya_Prescriptions

2026-07-12

Some simple models based on Tonmya sales

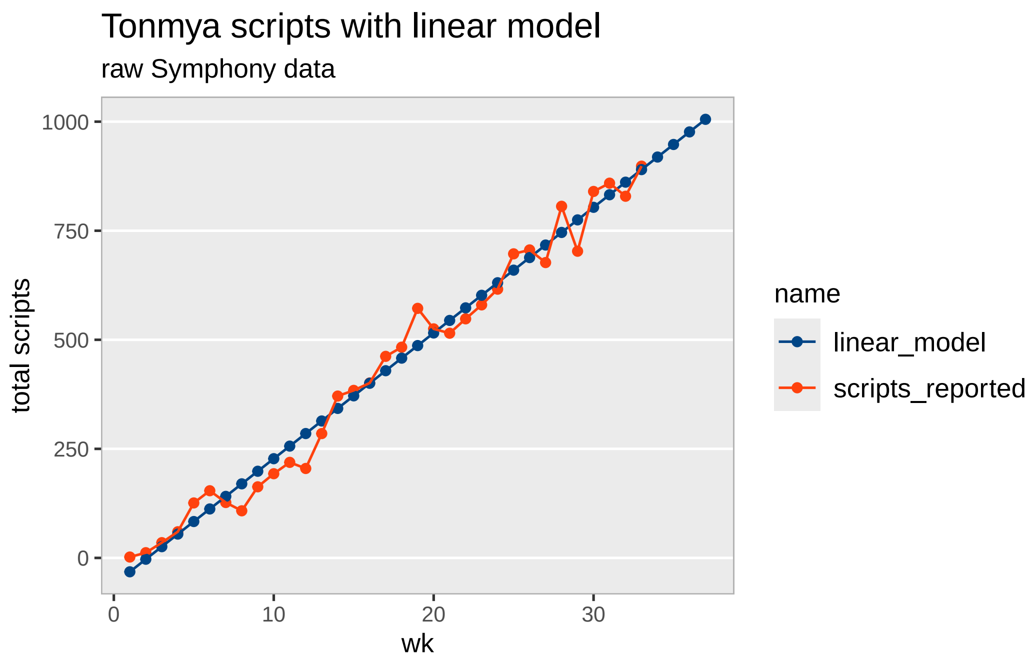
What are we doing here? - Just trying to get a grasp on the rate of increase and projecting out if that rate continues. It is not necessarily realistic to take it out as far as projections provide. Nothing here should be regarded as investment advice. I am not a financial analyst or advisor.

scripts Projection - at weekly rate - *linear model*

```
Call:
lm(formula = scripts ~ wk, data = df_scripts)

Residuals:
    Min      1Q  Median      3Q     Max
-80.072 -29.361   7.922  28.308  85.259

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -60.6458    13.9577  -4.345 0.000139 ***
wk           28.8098     0.7163  40.219  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39.18 on 31 degrees of freedom
Multiple R-squared:  0.9812,    Adjusted R-squared:  0.9806
F-statistic:  1618 on 1 and 31 DF,  p-value: < 2.2e-16
```

Scripts, Linear Model

wk	weekly_seq	total reported	linear predicted	annual
1	2025-11-14	2	-31.8	1,498
2	2025-11-21	12	-3.0	2,996
3	2025-11-28	35	25.8	4,494
4	2025-12-05	60	54.6	5,992
5	2025-12-12	126	83.4	7,491
6	2025-12-19	154	112.2	8,989
7	2025-12-26	127	141.0	10,487
8	2026-01-02	108	169.8	11,985
9	2026-01-09	163	198.6	13,483
10	2026-01-16	193	227.5	14,981
11	2026-01-23	219	256.3	16,479
12	2026-01-30	205	285.1	17,977
13	2026-02-06	285	313.9	19,475
14	2026-02-13	371	342.7	20,974
15	2026-02-20	384	371.5	22,472
16	2026-02-27	401	400.3	23,970
17	2026-03-06	462	429.1	25,468
18	2026-03-13	483	457.9	26,966
19	2026-03-20	572	486.7	28,464
20	2026-03-27	525	515.6	29,962
21	2026-04-03	515	544.4	31,460
22	2026-04-10	548	573.2	32,958
23	2026-04-17	580	602.0	34,457
24	2026-04-24	616	630.8	35,955
25	2026-05-01	697	659.6	37,453
26	2026-05-08	706	688.4	38,951
27	2026-05-15	677	717.2	40,449
28	2026-05-22	806	746.0	41,947
29	2026-05-29	703	774.8	43,445
30	2026-06-05	840	803.6	44,943
31	2026-06-12	859	832.5	46,441
32	2026-06-19	829	861.3	47,940
33	2026-06-26	898	890.1	49,438
34	2026-07-03	NA	918.9	50,936
35	2026-07-10	NA	947.7	52,434
36	2026-07-17	NA	976.5	53,932
37	2026-07-24	NA	1,005.3	55,430

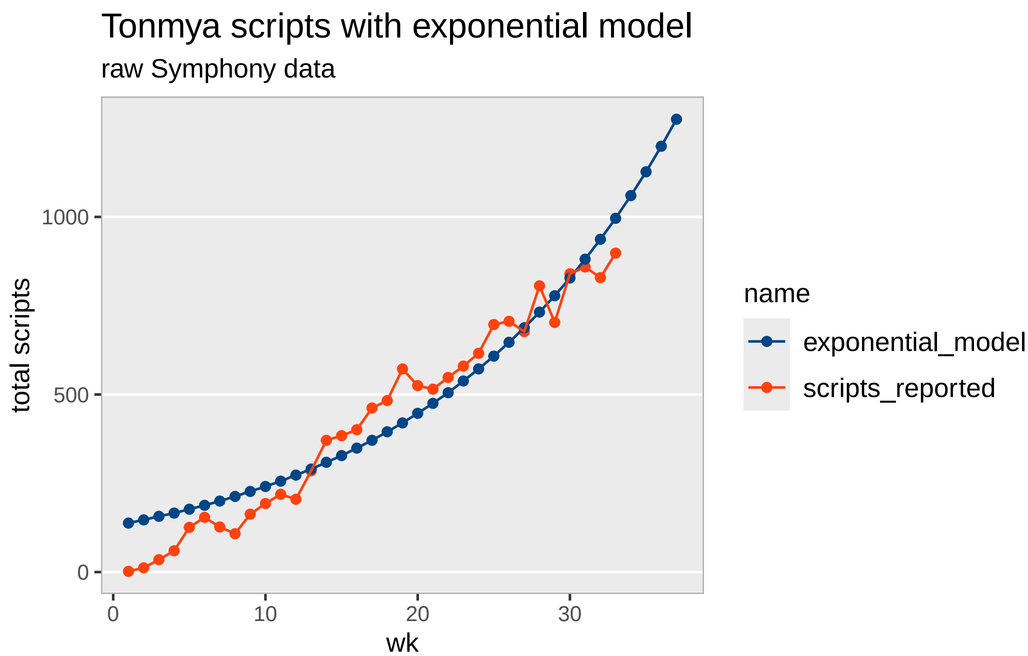


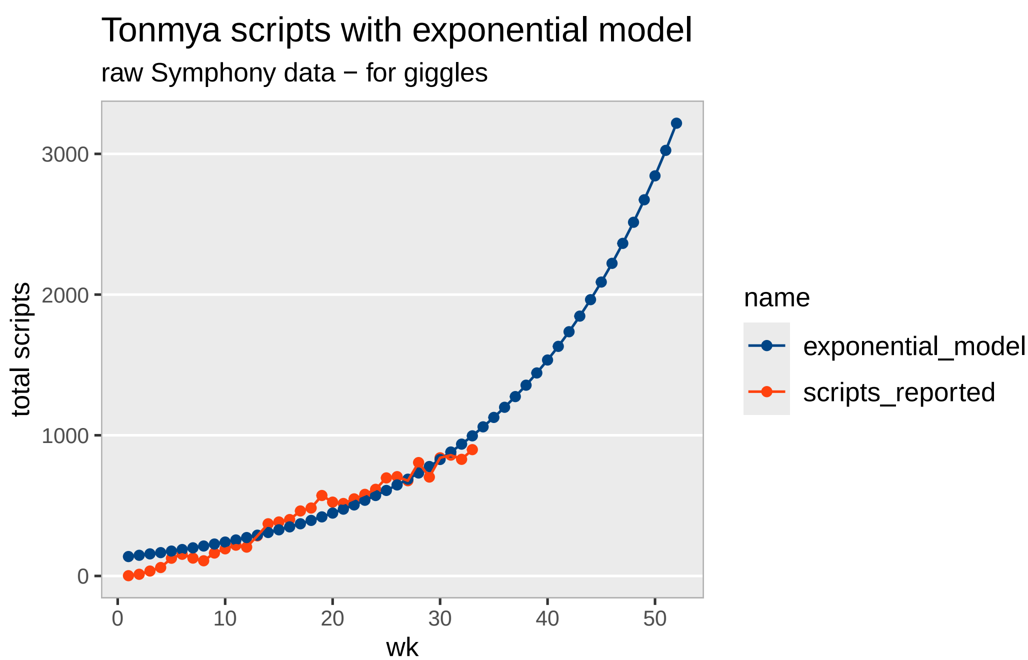
Scripts Projection - at weekly growth rate - *exponential model*

```
Formula: scripts ~ a * (1 + r)^wk

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 1.301e+02  1.428e+01    9.11 2.83e-10 ***
r 6.364e-02  4.351e-03   14.63 1.84e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 80.76 on 31 degrees of freedom

Number of iterations to convergence: 10
Achieved convergence tolerance: 5.507e-06
```

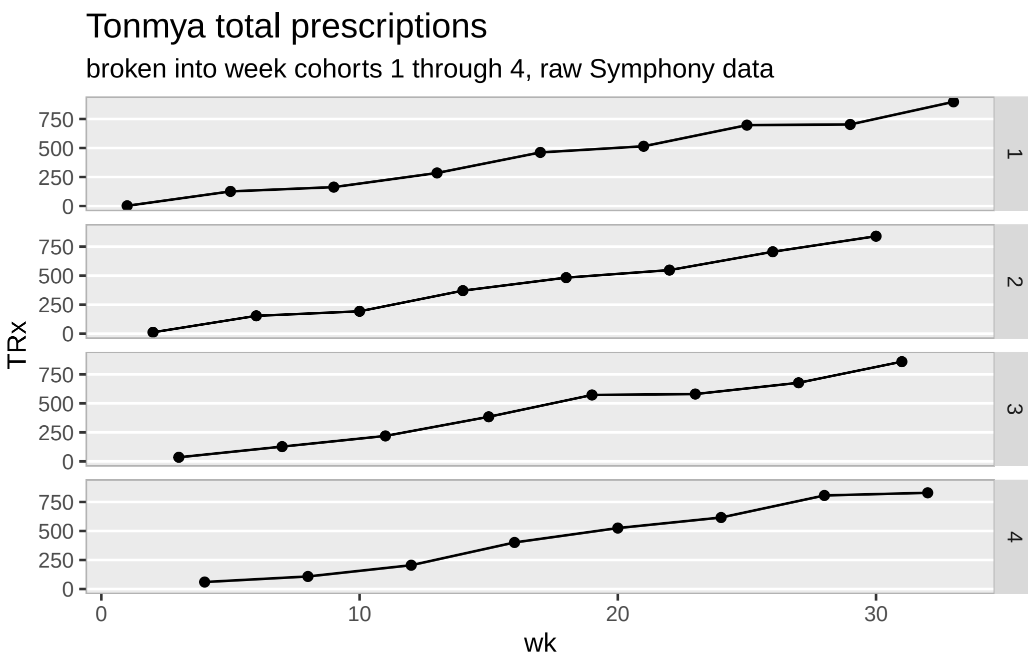
Scripts, Exponential Model

wk	weekly_seq	total reported	exp predicted	annual
1	2025-11-14	2	138	7,193
2	2025-11-21	12	147	7,651
3	2025-11-28	35	157	8,138
4	2025-12-05	60	166	8,656
5	2025-12-12	126	177	9,207
6	2025-12-19	154	188	9,793
7	2025-12-26	127	200	10,416
8	2026-01-02	108	213	11,079
9	2026-01-09	163	227	11,784
10	2026-01-16	193	241	12,534
11	2026-01-23	219	256	13,332
12	2026-01-30	205	273	14,181
13	2026-02-06	285	290	15,083
14	2026-02-13	371	309	16,043
15	2026-02-20	384	328	17,064
16	2026-02-27	401	349	18,150
17	2026-03-06	462	371	19,305
18	2026-03-13	483	395	20,534
19	2026-03-20	572	420	21,841
20	2026-03-27	525	447	23,231
21	2026-04-03	515	475	24,709
22	2026-04-10	548	505	26,282
23	2026-04-17	580	538	27,955
24	2026-04-24	616	572	29,734
25	2026-05-01	697	608	31,626
26	2026-05-08	706	647	33,639
27	2026-05-15	677	688	35,780
28	2026-05-22	806	732	38,057
29	2026-05-29	703	778	40,479
30	2026-06-05	840	828	43,056
31	2026-06-12	859	881	45,796
32	2026-06-19	829	937	48,711
33	2026-06-26	898	996	51,811
34	2026-07-03	NA	1060	55,108
35	2026-07-10	NA	1127	58,616
36	2026-07-17	NA	1199	62,346
37	2026-07-24	NA	1275	66,314





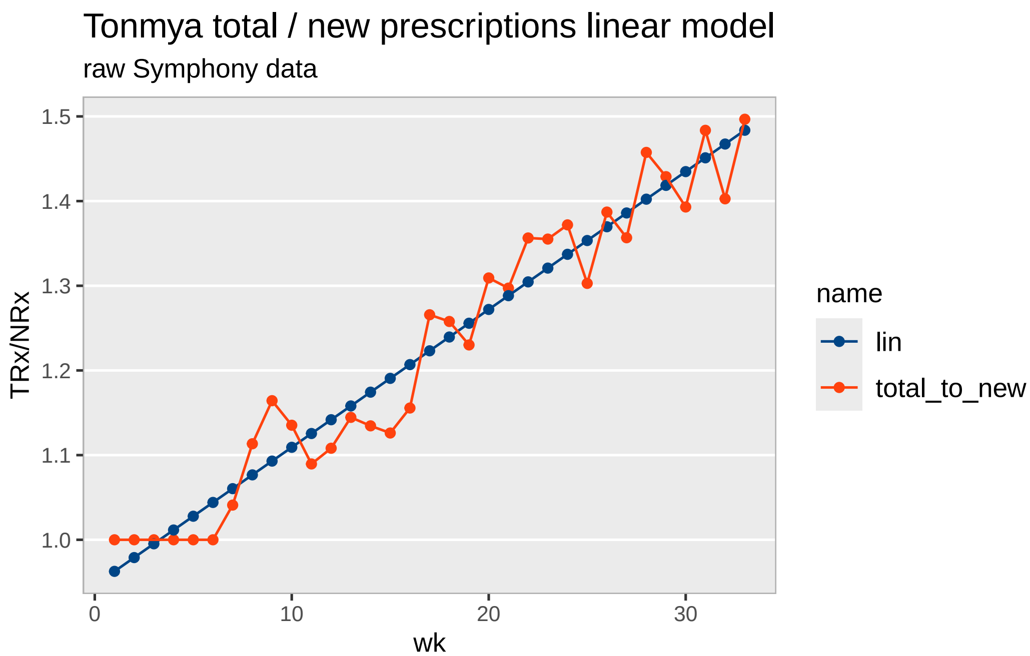
TRx Total Prescriptions by Weekly Cohorts



total / new scripts at weekly rate - *linear model*

```
Call:
lm(formula = total_to_new ~ wk, data = trx_nrx_df)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.064680 -0.033674  0.008925  0.034276  0.071345 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.9464187  0.0138058   68.55   <2e-16 ***
wk          0.0162803  0.0007085   22.98   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03876 on 31 degrees of freedom
Multiple R-squared:  0.9445,    Adjusted R-squared:  0.9428 
F-statistic:   528 on 1 and 31 DF,  p-value: < 2.2e-16
```

wk	scripts_reported
1	
1	2
5	126
9	163
13	285
17	462
21	515
25	697
29	703
33	898
2	
2	12
6	154
10	193
14	371
18	483
22	548
26	706
30	840
3	
3	35
7	127
11	219
15	384
19	572
23	580
27	677
31	859
4	
4	60
8	108
12	205
16	401
20	525
24	616
28	806
32	829

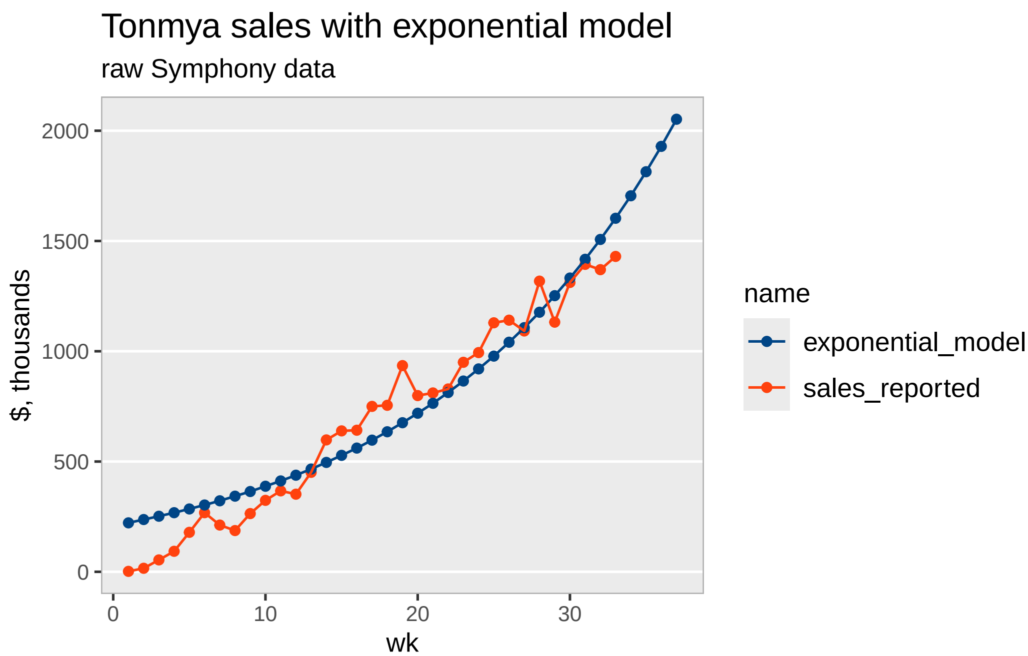


Sales Projection - at weekly growth rate - *exponential model*

```
Formula: sales ~ a * (1 + r)^wk

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 2.090e+02  2.271e+01   9.203 2.24e-10 ***
r 6.368e-02  4.306e-03  14.788 1.37e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 128.6 on 31 degrees of freedom

Number of iterations to convergence: 13
Achieved convergence tolerance: 5.346e-06
```

Exponential Model

wk	weekly_seq	sales, thousands	exp, thousands	annual, M
1	2025-11-14	$2	$222	$11.6
2	2025-11-21	$16	$237	$12.3
3	2025-11-28	$54	$252	$13.1
4	2025-12-05	$93	$268	$13.9
5	2025-12-12	$179	$285	$14.8
6	2025-12-19	$268	$303	$15.7
7	2025-12-26	$212	$322	$16.7
8	2026-01-02	$187	$343	$17.8
9	2026-01-09	$264	$364	$18.9
10	2026-01-16	$324	$388	$20.2
11	2026-01-23	$367	$412	$21.4
12	2026-01-30	$352	$438	$22.8
13	2026-02-06	$451	$466	$24.3
14	2026-02-13	$598	$496	$25.8
15	2026-02-20	$639	$528	$27.4
16	2026-02-27	$642	$561	$29.2
17	2026-03-06	$750	$597	$31.0
18	2026-03-13	$755	$635	$33.0
19	2026-03-20	$935	$676	$35.1
20	2026-03-27	$799	$719	$37.4
21	2026-04-03	$811	$764	$39.7
22	2026-04-10	$829	$813	$42.3
23	2026-04-17	$950	$865	$45.0
24	2026-04-24	$994	$920	$47.8
25	2026-05-01	$1,129	$978	$50.9
26	2026-05-08	$1,141	$1,041	$54.1
27	2026-05-15	$1,092	$1,107	$57.6
28	2026-05-22	$1,318	$1,177	$61.2
29	2026-05-29	$1,132	$1,252	$65.1
30	2026-06-05	$1,312	$1,332	$69.3
31	2026-06-12	$1,394	$1,417	$73.7
32	2026-06-19	$1,370	$1,507	$78.4
33	2026-06-26	$1,430	$1,603	$83.4
34	2026-07-03	NA	$1,705	$88.7
35	2026-07-10	NA	$1,814	$94.3
36	2026-07-17	NA	$1,929	$100.3
37	2026-07-24	NA	$2,052	$106.7


Tonmya sales with exponential model
raw Symphony data
$, thousands
2000
1500
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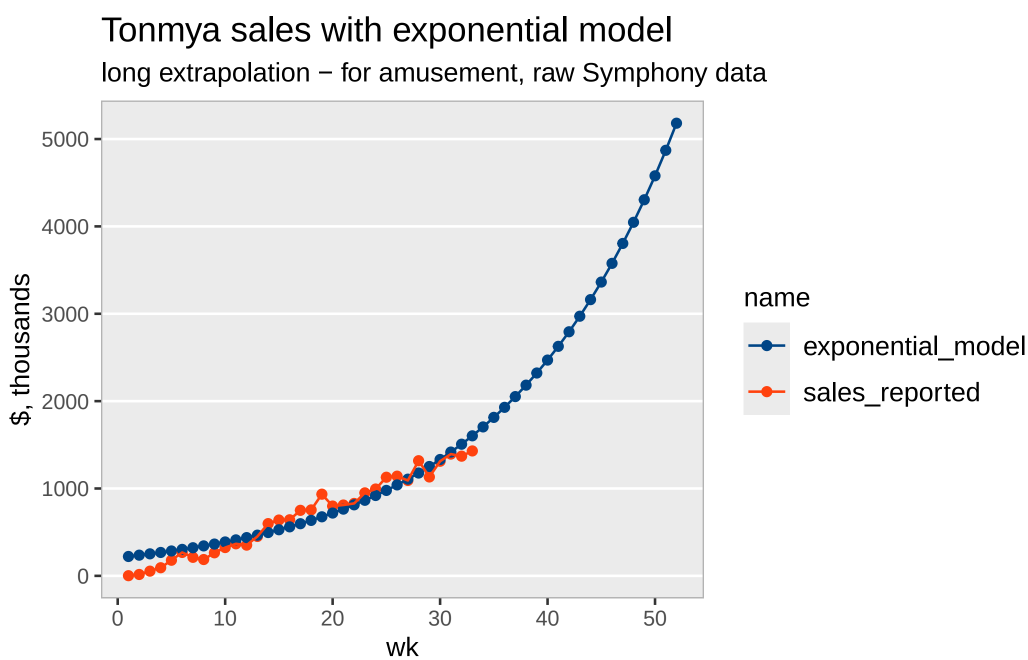
30
wk
name
exponential_model
sales_reported


Tonmya sales with exponential model
long extrapolation − for amusement, raw Symphony data
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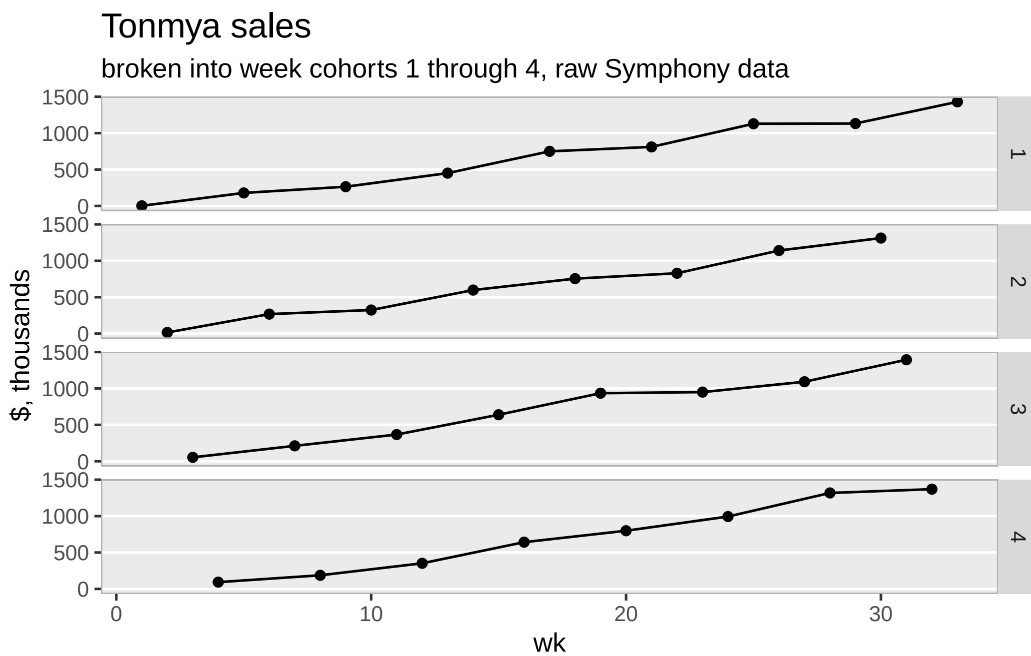
name
exponential_model
sales_reported

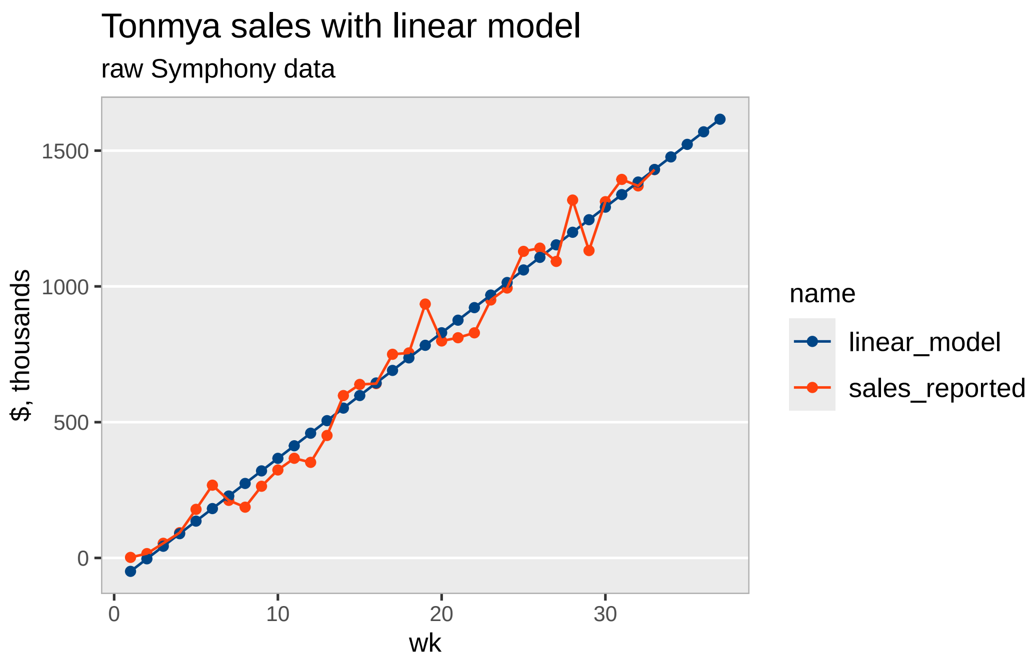


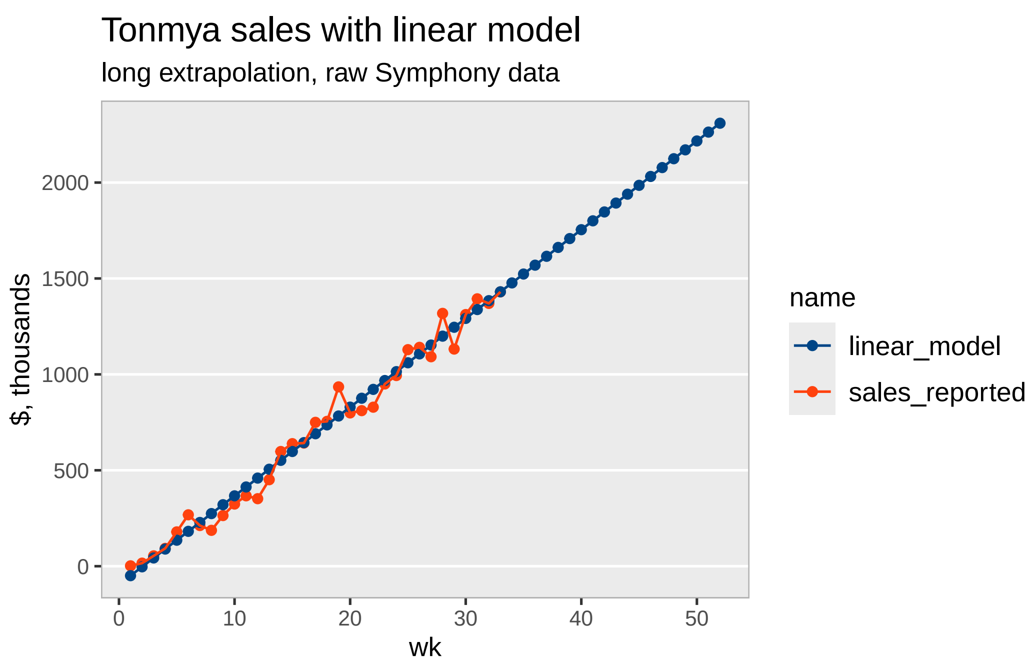
Sales Projection - at weekly rate - *linear model*

```
Call:
lm(formula = sales ~ wk, data = df_sales)

Residuals:
     Min       1Q   Median       3Q      Max
-113.568  -46.080   -0.565   43.416  151.926

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -95.663     22.845  -4.187 0.000216 ***
wk            46.249      1.172  39.447  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 64.13 on 31 degrees of freedom
Multiple R-squared:  0.9805,    Adjusted R-squared:  0.9798
F-statistic:  1556 on 1 and 31 DF,  p-value: < 2.2e-16
```

Linear Model

wk	weekly_seq	sales, thousands	linear, thousands	annual, M
1	2025-11-14	$2	-$49	$2.4
2	2025-11-21	$16	-$3	$4.8
3	2025-11-28	$54	$43	$7.2
4	2025-12-05	$93	$89	$9.6
5	2025-12-12	$179	$136	$12.0
6	2025-12-19	$268	$182	$14.4
7	2025-12-26	$212	$228	$16.8
8	2026-01-02	$187	$274	$19.2
9	2026-01-09	$264	$321	$21.6
10	2026-01-16	$324	$367	$24.0
11	2026-01-23	$367	$413	$26.5
12	2026-01-30	$352	$459	$28.9
13	2026-02-06	$451	$506	$31.3
14	2026-02-13	$598	$552	$33.7
15	2026-02-20	$639	$598	$36.1
16	2026-02-27	$642	$644	$38.5
17	2026-03-06	$750	$691	$40.9
18	2026-03-13	$755	$737	$43.3
19	2026-03-20	$935	$783	$45.7
20	2026-03-27	$799	$829	$48.1
21	2026-04-03	$811	$876	$50.5
22	2026-04-10	$829	$922	$52.9
23	2026-04-17	$950	$968	$55.3
24	2026-04-24	$994	$1,014	$57.7
25	2026-05-01	$1,129	$1,061	$60.1
26	2026-05-08	$1,141	$1,107	$62.5
27	2026-05-15	$1,092	$1,153	$64.9
28	2026-05-22	$1,318	$1,199	$67.3
29	2026-05-29	$1,132	$1,246	$69.7
30	2026-06-05	$1,312	$1,292	$72.1
31	2026-06-12	$1,394	$1,338	$74.6
32	2026-06-19	$1,370	$1,384	$77.0
33	2026-06-26	$1,430	$1,431	$79.4
34	2026-07-03	NA	$1,477	$81.8
35	2026-07-10	NA	$1,523	$84.2
36	2026-07-17	NA	$1,569	$86.6
37	2026-07-24	NA	$1,616	$89.0





Refills - at weekly rate - *linear model*

```
Call:
lm(formula = refills ~ wk, data = trx_nrx_df)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-39.236 -14.965  -4.546  12.493  50.385 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -59.8580     9.1738  -6.525 2.78e-07 ***
wk            9.4729     0.4708  20.120  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 25.75 on 31 degrees of freedom
Multiple R-squared:  0.9289,	Adjusted R-squared:  0.9266 
F-statistic: 404.8 on 1 and 31 DF,  p-value: < 2.2e-16
```